

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\langle \tilde{v}_\lambda | B(y_n | t) \cdots B(y_1 | t) v_{0 \dots 0} \rangle = \frac{\Pi(y)}{\Pi(t_\lambda)} \sum_{\mathcal{T} \in \text{SVT}(\lambda^\vee)} \beta^{|\mathcal{T}| - |\lambda^\vee|} \prod_{\substack{\langle i, j \rangle \in \lambda^\vee \\ r \in \mathcal{T}(i, j)}} y_r \ominus t'_{r+j-i}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$I - \mathfrak{z}_1 T_{1,0}^* - \mathfrak{z}_2 T_{2,0}^* = \begin{pmatrix} 1 & \frac{1}{2}(\mathfrak{z}_1 + \mathfrak{z}_2) & \frac{1}{2}(\mathfrak{z}_1 + \mathfrak{z}_2) \\ 0 & 1 + \frac{1}{2}\mathfrak{z}_2 & -\frac{1}{2}\mathfrak{z}_2 \\ 0 & -\frac{1}{2}\mathfrak{z}_1 & 1 + \frac{1}{2}\mathfrak{z}_1 \end{pmatrix}.$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \sum_{r=1}^{a-1} \left(\sum_{i=1}^{k-1} x_i \right)_r &= m \left(\sum_{i=0}^{k-1} i \binom{k-1}{i} + \sum_{j=1}^{k-1} \left(\sum_{i=0}^{k-j-1} (i+2) \binom{k-j-1}{i} \right) \right) \\ &\quad + \left(\sum_{i=0}^{k-2} i \binom{k-2}{i} + \sum_{j=1}^{k-2} \left(\sum_{i=0}^{k-j-2} (i+2) \binom{k-j-2}{i} \right) \right) \\ &= m(2^{k-1}k - 1) + 2^{k-2}(k-1) - 1. \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{pmatrix} p'_0 \\ p'_1 \end{pmatrix} = \frac{1}{\sqrt{1 - B_{01}^2}} \begin{pmatrix} 1 & 0 \\ -B_{01} & 1 - B_{01}^2 \end{pmatrix} \begin{pmatrix} p_0 - B_{01}p_1 \\ p_1 \end{pmatrix} = \begin{pmatrix} \gamma p_0 - \gamma v p_1 \\ -v \gamma p_0 + \gamma p_1 \end{pmatrix}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} g_{rr} &= \frac{l^2}{r^2} + O(r^{-4}) & g_{tt} &= -\frac{r^2}{l^2} + O(1) \\ \textit{beginalign*2mm} g_{tr} &= O(r^{-3}) & g_{\varphi\varphi} &= r^2 + O(1) \\ \textit{beginalign*1mm} g_{\varphi r} &= O(r^{-3}) & g_{t\varphi} &= O(1) \end{aligned}$$